

# Integrals – 2

## Symmetry, the king's rule and periodicity

### Homework

1 сентября 2026 года

**Theoretical minimum.** The king's rule  $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$  and the trick of adding an integral to itself; parity; periodicity and the integral over a period; the substitutions  $x \mapsto 1/x$  and  $x \mapsto a/x$ ; the identity  $\int_{-a}^a \frac{f}{1+e^x} = \int_0^a f$  for even  $f$ ; orthogonality of  $\{\sin nx, \cos nx\}$ .

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1. Prove that  $\int_{-a}^a \frac{f(x)}{1+e^x} dx = \int_0^a f(x) dx$  for even  $f$ , and apply it to  $\int_{-1}^1 \frac{x^2}{1+e^x} dx$  and  $\int_{-\pi/2}^{\pi/2} \frac{\cos x}{1+e^x} dx$ .

2. (Integration Bee 2022) Compute

$$\int \left( \frac{1}{1+\sin x} + \frac{1}{1+\cos x} + \frac{1}{1+\tan x} + \frac{1}{1+\cot x} + \frac{1}{1+\sec x} + \frac{1}{1+\csc x} \right) dx.$$

3. (Integration Bee 2023) Compute  $\int_0^{2\pi} [2023 \sin x] dx$ .

4. (Integration Bee) Nested absolute values:  $\int_{-1}^1 |x+|x+\dots+|x+|x|| \dots| dx$  with 2026 occurrences of  $x$  (2026), and  $\int_{-2023}^{2023} |\dots||x|-1|-1 \dots| dx$  with 2023 subtractions (2023).

5. (Integration Bee) Symmetry of the interval  $[0, 1]$ :  $\int_0^1 x^{2024}(1-x^{2025})^{2025} dx$  (2025) and  $\int x(1-x)^{2020} dx$  (2020); explain why in the first one the substitution  $x^{2025} \mapsto 1-x^{2025}$  is convenient.

6. (Integration Bee 2025) Compute  $\int \left( \cos x + \cos\left(x + \frac{2\pi}{3}\right) + \cos\left(x - \frac{2\pi}{3}\right) \right)^2 dx$  and explain the answer through the sum of the cube roots of unity.

7. Suppose  $f(a+b-x) = g(x)$  on  $[a, b]$  and  $f, g > 0$ . Prove that  $\int_a^b \frac{f}{f+g} = \frac{b-a}{2}$ , and construct an example where this gives the answer although the antiderivative is not elementary.

8. (Integration Bee 2026) Compute  $\int_0^{1000} (\lfloor \lceil x \rceil \rfloor + \lceil \lfloor x \rfloor \rceil + \lfloor \{x\} \rfloor + \{ \lfloor x \rfloor \} + \lceil \{x\} \rceil + \{ \lceil x \rceil \}) dx$ , treating each summand separately.
9. (Integration Bee) Step binomial coefficients:  $\int_0^{20} \binom{\lfloor x \rfloor}{2} dx$  (2025) and  $\int_0^{2026} \binom{\lfloor x \rfloor}{3} dx$  (2026). Reduce both to sums of binomial coefficients and use the hockey-stick identity.
10. Prove that  $\int_0^{2\pi} f(\sin x) \cos x \, dx = 0$  for every continuous  $f$ , and state the general criterion: if the substitution  $x \mapsto c-x$  or  $x \mapsto x+T$  changes the sign of the integrand, the integral vanishes. Test the criterion on three Integration Bee integrals of your choice from the class sheet.